

The Lifetime Incidence of Transfer Programs

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Motivation

How targeted is the U.S. safety net at those in greatest need?

- Current income measures consumption imperfectly, esp. at bottom of distribution
 - Lifetime incidence may be a more informative measure of distributional impact (Modigliani & Brumberg 1954, Friedman 1957, Meyer & Sullivan 2003)
- Mobility suggests lifetime benefits are less progressive than current benefits
 - Due to mobility, many individuals are eligible for transfers for part of their lifetimes
- Yet both eligibility rules & take-up behavior affect incidence of transfers
 - Contrast with lifetime tax burdens (Poterba 1989/1991, Fullerton & Lim 1993)

This Paper

① Estimate lifetime incidence of eight means-tested transfer programs

- Lifetime incidence is *more* progressive than current incidence (exception: UI)
- In total: \$4500 vs. \$2200 per person-year at bottom of lifetime vs. current distribution
- Total per-capita benefits decline four times faster in lifetime rank than in current rank

② Decompose Δ of lifetime vs. current incidence: mobility, eligibility, take-up

- Mobility effect makes lifetime incidence more uniform than current incidence, but it is overwhelmed by eligibility and take-up effects
- Eligibility rules & selective take-up both improve targeting: Lifetime earnings ranks of program participants are lower than nonparticipants with same current earnings

Literature

- **Lifetime incidence of taxes and government policies**

Davies et al. 1984, Poterba 1989/1991, Slemrod 1992, Fullerton & Lim 1993, Lyon & Schwab 1995, Bengtsson et al. 2016, Catherine & Yannelis 2020

→ first analysis of lifetime incidence with realistic heterogeneity or incomplete take-up

- **Targeting properties of transfer programs**

Akerlof 1978, Nichols & Zeckhauser 1982, Parsons 1991, Diamond & Sheshinski 1995, Dynarski & Scott-Clayton 2006, Kleven & Kopczuk 2011, Bhargava & Manoli 2015, Alatas et al. 2016, Deshpande & Li 2019, Finkelstein & Notowidigdo 2019, Gray 2019, Stepner 2019, Hendren 2020

→ new decomposition offers a precise & generalizable meaning to claims that eligibility rules or selective take-up improve or harm targeting

Outline

- ① Data and Measurement
- ② Current and Lifetime Incidence
- ③ Decomposition
- ④ Selection by Eligibility and Take-Up

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Data Overview

Panel Survey of Income Dynamics (PSID) waves, 1997–2019

- *Earnings & Consumption:*
 - Track people from ages 18 to 65 if household head or spouse
 - High-quality panel, used extensively in research on lifetime tax burdens
- *Program Participation & Benefit Levels:*
 - Can observe SNAP, Medicaid, SSI, TANF, UI, WIC, LIHEAP, public housing, Section 8
 - Currently validating results using SIPP linked to admin data

Other Data Sources

- Simulated eligibility using PSID + program rules
- CPS: Impute cash-equivalent values of Medicaid & housing assistance

Estimating Lifetime Earnings

We estimate a panel data model of individual earnings by Poisson PML:

$$\log E[y_{it} | x_{it}] = \alpha_i + \alpha_t + x'_{it}\beta$$

where

- α_i : individual FE
- α_t : year FE
- x_{it} : time-varying covariates (age, marital status, children, education)

Steps to measure lifetime earnings

[Details](#)[Validation](#)

- 1 Shrink estimated FEs $\{\hat{\alpha}_i\}$ by Empirical Bayes
- 2 Fill in panel by imputing missing x_{it} and using predicted values of y_{it}
- 3 For households: Add up predicted earnings and equalize

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Who Benefits from Means-Tested Transfers?

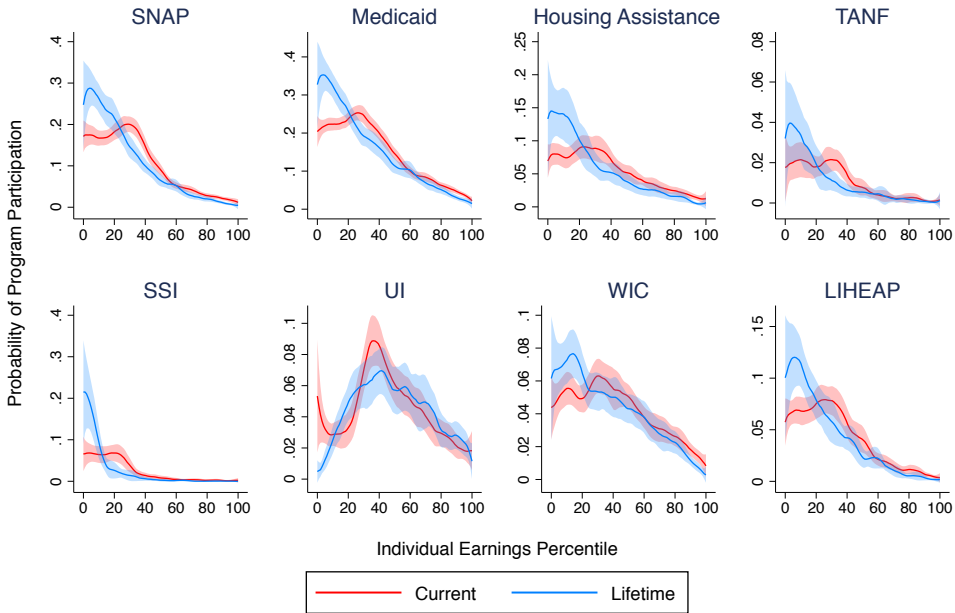
We study incidence graphically by local linear regressions of outcomes on earnings ranks.

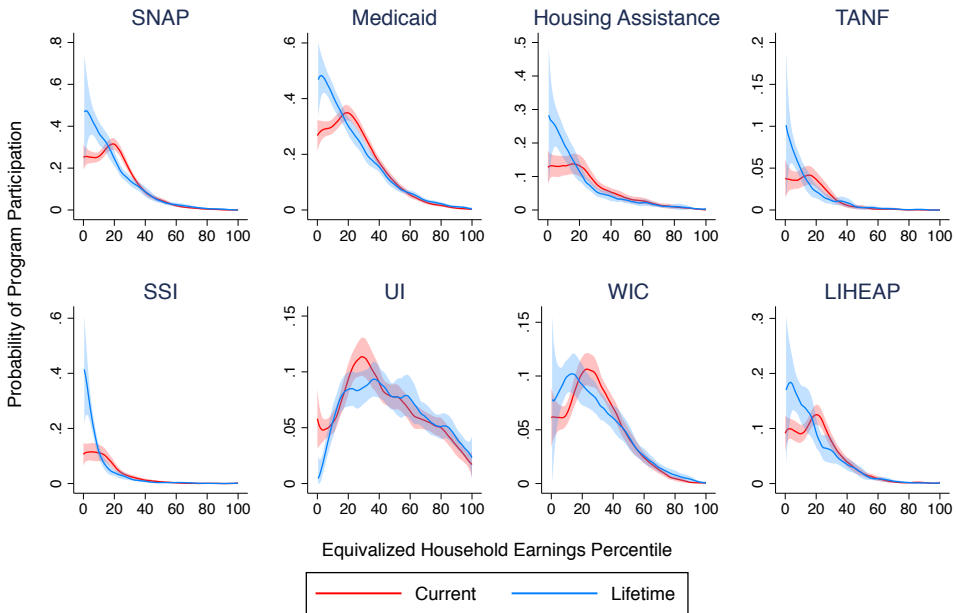
Outcomes:

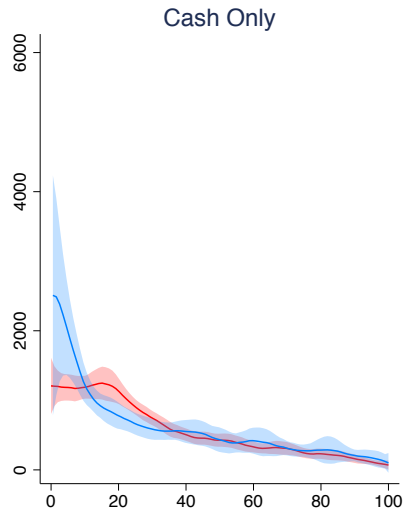
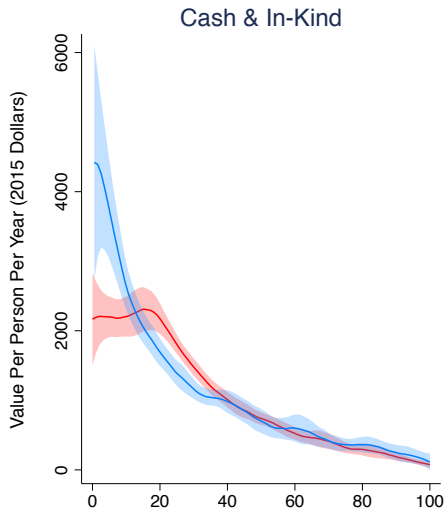
- Program participation rates
- Annual per-capita value of benefits

Throughout looking at current and lifetime earnings ranks of:

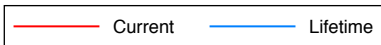
- Individuals
- Equivalized households
- Unequivalized households



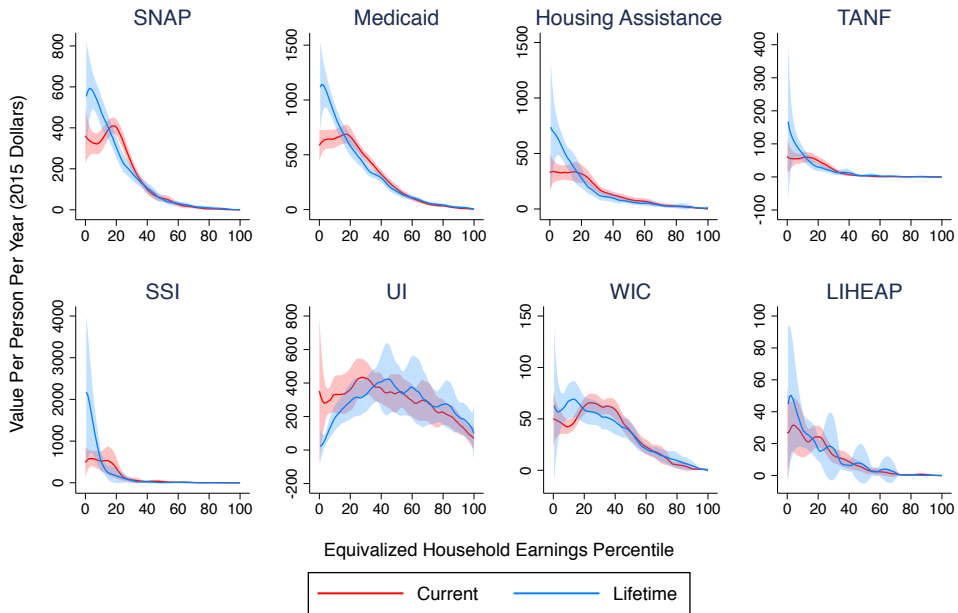




Equivalized Household Earnings Percentile



More



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Mobility, Eligibility, and Take-Up Effects

Why is lifetime incidence more progressive than current incidence?

3 forces to explain difference in incidence:

- ① **Mobility:** Life-cycle trends or transitory shocks cause people to change current earnings ranks over their lifetimes
- ② **Eligibility:** Screening criteria other than current earnings may have residual predictive power for lifetime earnings
- ③ **Take-Up:** Among eligibles, take-up probability depends on lifetime earnings rank, not only current earnings rank

→ Can be applied to program participation and per-capita annual value of transfers

Decomposition

Let $D_{it} = 1$ indicate participation by i in t . By the law of conditional expectation:

$$\Pr(D_{it} = 1 \mid \bar{r}_i = r) = \int \Pr(D_{it} = 1 \mid r_{it}, \bar{r}_i = r) dF(r_{it} \mid \bar{r}_i = r).$$

We use this to define two participation counterfactuals.

Counterfactual 1 (mobility effect only): Lifetime incidence if participation were purely a function of current earnings rank, then run through lifetime mobility matrix

$$\widetilde{p}_M(r) = \int \Pr(D_{it} = 1 \mid r_{it}) dF(r_{it} \mid \bar{r}_i = r)$$

- $\Pr(D_{it} = 1 \mid r_{it})$: participation rate at current rank r_{it}
- $F(r_{it} \mid \bar{r}_i = r)$: conditional distribution of current ranks r (given lifetime rank $\bar{r}_i$)

Decomposition

Let $D_{it} = 1$ indicate participation by i in t . By the law of conditional expectation:

$$\Pr(D_{it} = 1 \mid \bar{r}_i = r) = \int \Pr(D_{it} = 1 \mid r_{it}, \bar{r}_i = r) dF(r_{it} \mid \bar{r}_i = r).$$

We use this to define two participation counterfactuals.

Counterfactual 2 (mobility + eligibility effects): Lifetime rank matters only via eligibility, thus no selective take-up among eligibles on lifetime rank

$$\widetilde{p}_E(r) = \int \Pr(D_{it} = 1 \mid r_{it}, E_{it} = 1) \Pr(E_{it} = 1 \mid r_{it}, \bar{r}_i = r) dF(r_{it} \mid \bar{r}_i = r)$$

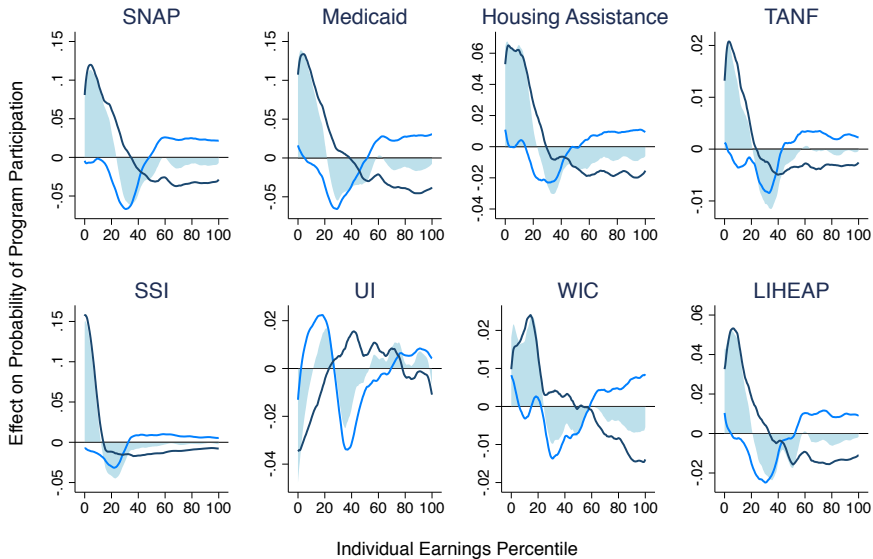
- $\Pr(D_{it} = 1 \mid E_{it} = 1, r_{it})$: take-up rate among eligibles at current rank r_{it}
- $\Pr(E_{it} = 1 \mid r_{it}, \bar{r}_i = r)$: eligible share given current rank r_{it} and lifetime rank $\bar{r}_i$

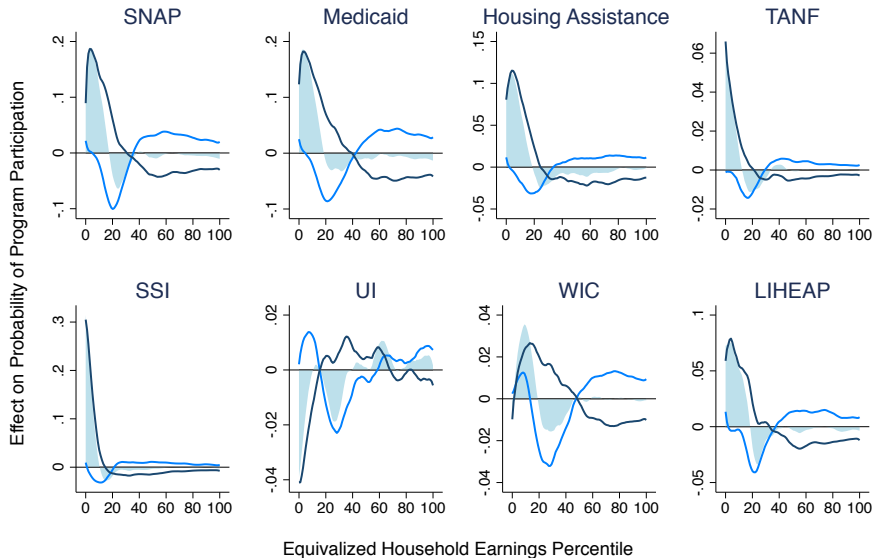
Decomposition

Let $p_L(r) = \Pr(D_{it} = 1 \mid \bar{r}_i = r)$ and $p_C(r) = \Pr(D_{it} = 1 \mid r_{it} = r)$. The decomposition is:

$$p_L(r) - p_C(r) = \underbrace{[p_L(r) - \widetilde{p}_E(r)]}_{\text{take-up effect}} + \underbrace{[\widetilde{p}_E(r) - \widetilde{p}_M(r)]}_{\text{eligibility effect}} + \underbrace{[\widetilde{p}_M(r) - p_C(r)]}_{\text{mobility effect}}.$$

- **Mobility effect:** Current incidence vs. lifetime incidence in counterfactual where participation is uninformative about lifetime earnings given current earnings
- **Eligibility effect:** Comparison of lifetime incidence in two counterfactuals: participation is uninformative vs. participation is informative only via eligibility
- **Take-up effect:** Actual lifetime incidence vs. lifetime incidence in counterfactual in which participation is informative only via eligibility



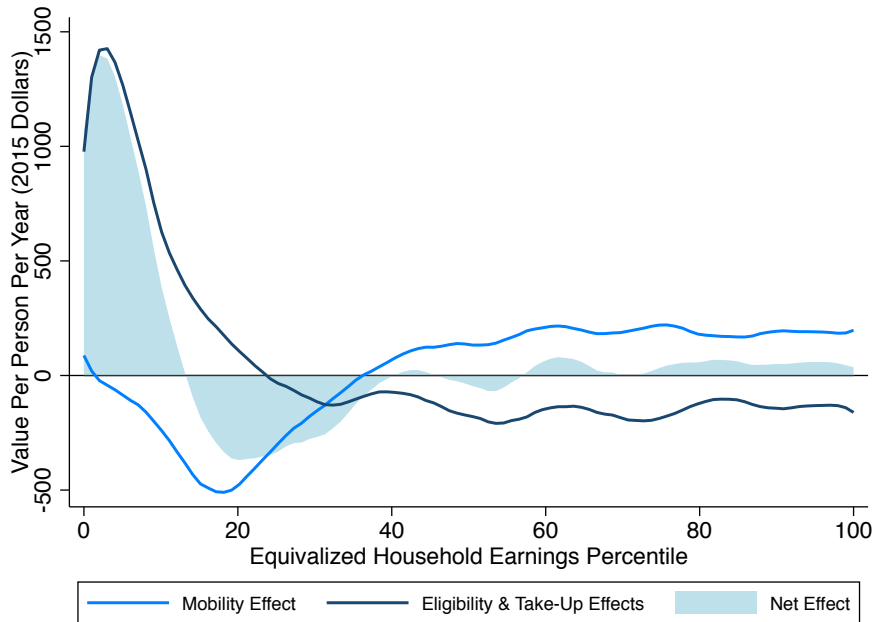


— Mobility Effect

— Eligibility & Take-Up Effects

Net Effect

More

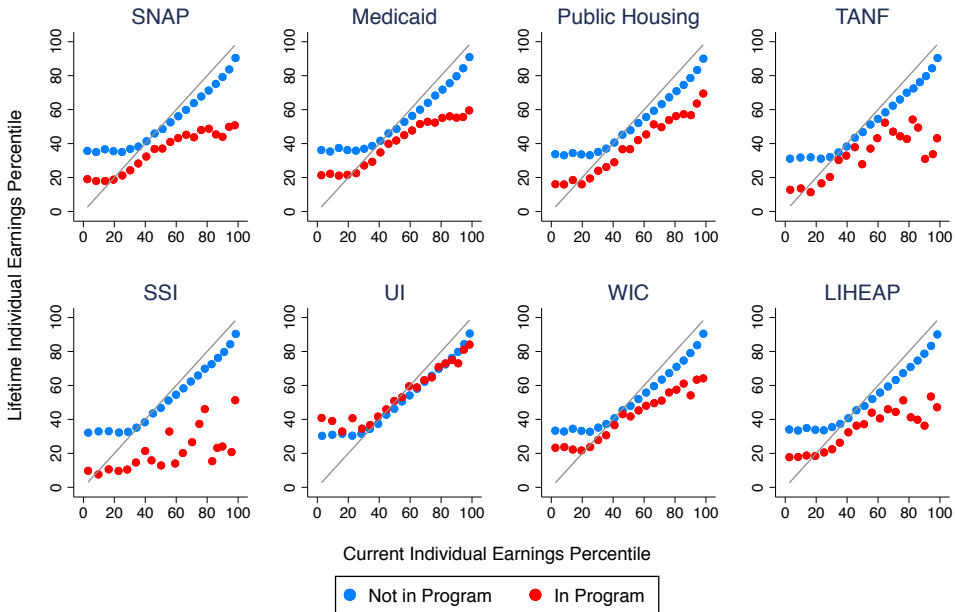


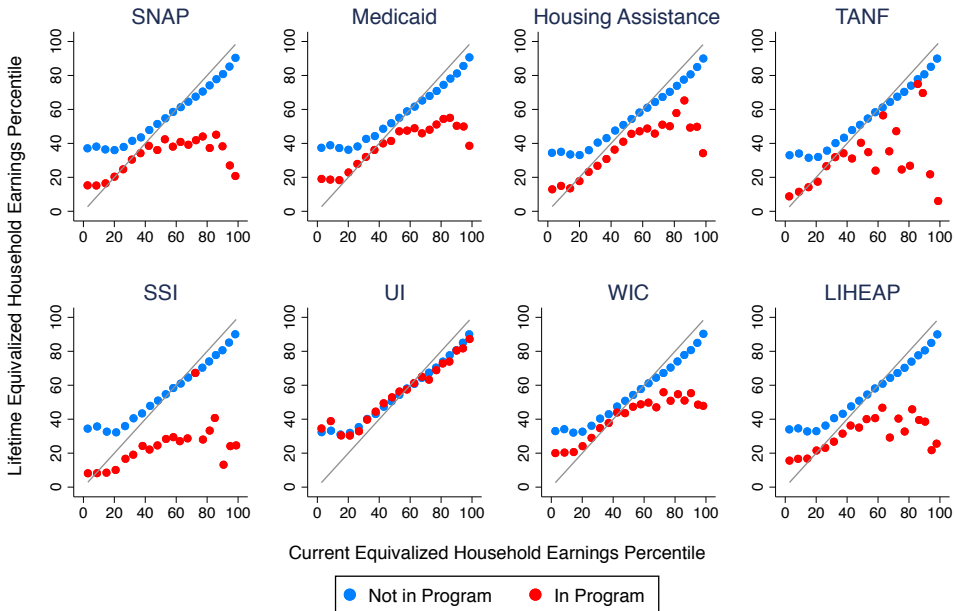
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Mechanisms

- So far, only indirect evidence for eligibility & take-up forces: actual and counterfactual lifetime incidence are highly dissimilar
- Now, direct evidence: Lifetime earnings ranks of program participants are consistently lower than nonparticipants at the same earnings rank
- To do [for PF lunch]: Understand selection into participation on lifetime earnings
 - How much is explained by eligibility?
 - Among eligibles, how much is explained by characteristics the government can observe?





Conclusion

- Distributional analysis with respect to current earnings gives a systematically misleading view of the U.S. safety net
 - But opposite in direction from bias identified by lifetime tax burden literature
 - More highly targeted at neediest: lifetime poorest and lowest in consumption
- Progressive lifetime incidence reflects both eligibility rules and selective take-up
 - Both strongly select on lifetime earnings rank conditional on current earnings rank
 - (to come pending results)

Backup Slides

Step 1: Empirical Bayes Shrinkage of Estimated Fixed Effects

Following Morris (1983), we shrink estimated FEs towards conditional mean predicted by time-invariant covariates (Jacobs & Lefgren 2007, Chandra et al. 2016, Sorkin 2017)

$$\alpha_i^* = \frac{\sigma_i^2}{\sigma_i^2 + \tau^2} \cdot z_i' \lambda + \frac{\tau^2}{\sigma_i^2 + \tau^2} \cdot \hat{\alpha}_i$$

where

- z_i : time-invariant covariates (sex, race)
- σ_i^2 : sampling variance of $\hat{\alpha}_i$ (sample analog obtained via bootstrap)
- τ^2 : variance of true FEs α_i

→ Procedure iteratively updates $\hat{\lambda}, \hat{\tau}$ until convergence

Steps 2 and 3: Fill in Panel & Equivalize Household Earnings

Step 2. Impute covariates & predicted earnings if not observed at ages from 18 to 65

→ Assume that x_{it} unchanged at latest observed value

$$\widehat{y}_{it}^* = \exp \left(\widehat{\alpha}_i^* + \widehat{\alpha}_{ct} + x_{it} \widehat{\beta} \right)$$

Step 3. For households, add up earnings and equivalize:

$$\widehat{y}_{ht}^* = f \left(\widehat{y}_{i(h)t}^* + \widehat{y}_{j(h)t}^* \right)$$

where

- $i(h)$ and $j(h)$: head and spouse of household h
- $f(\cdot)$: equivalence scale function (Meyer & Sullivan 2008)

Equivalence Scale Details

$$y_{ht} = \frac{1}{\psi_h} \sum_{i \in h} y_{it}$$

Options:

- Two-parameter family

$$\psi_h = (N_h^{\text{adults}} + \beta \cdot N_h^{\text{children}})^{\gamma}$$

$$\beta = 0.7, \gamma = 0.7$$

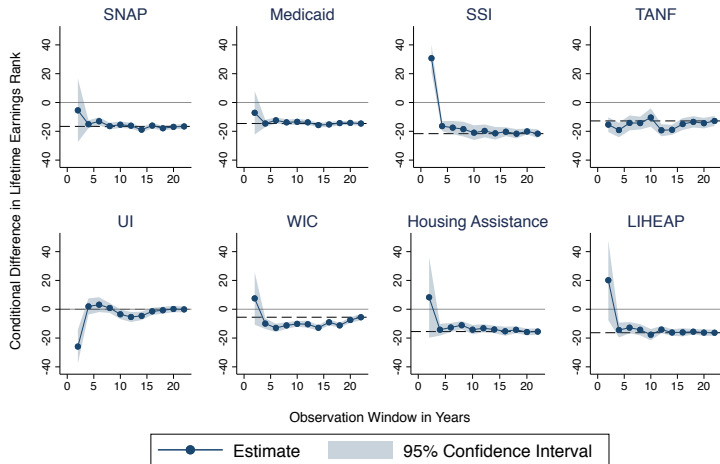
- Square-root

$$\psi_h = (N_h^{\text{adults}} + N_h^{\text{children}})^{1/2}$$

- OECD

$$\psi_h = (1 + 0.5 \cdot (N_h^{\text{adults}} - 1) + 0.3 \cdot N_h^{\text{children}})$$

Stable Predictive Effect of Transfers on Lifetime Rank



$$\bar{r}_i = \alpha + \beta r_{it} + \gamma D_{it} + e_i$$

Earnings Percentile Rank Definitions

Individual current earnings rank

$$r_{it} = \Pr(y \leq y_{it})$$

Household current earnings rank

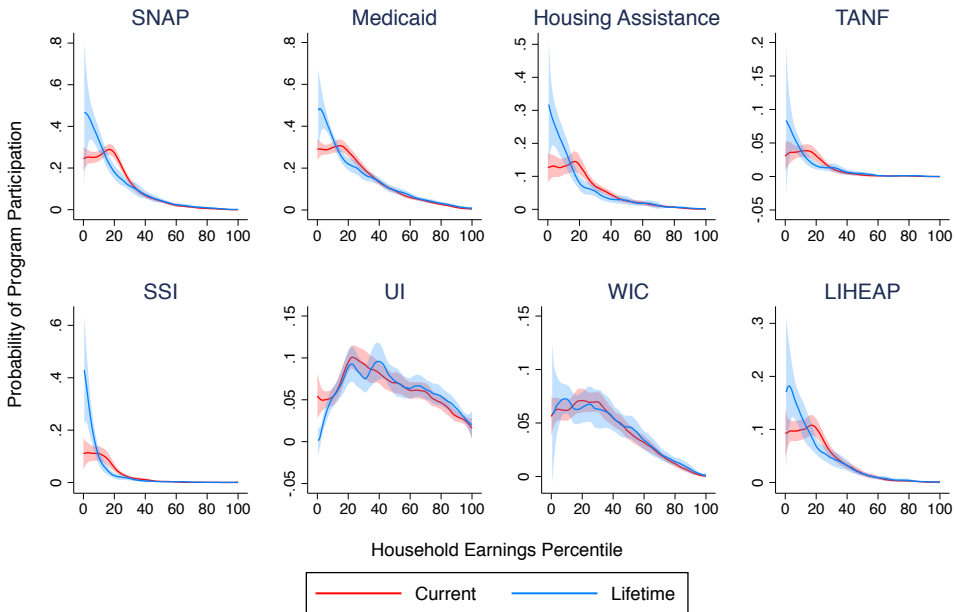
$$r_{ht} = \Pr(y \leq y_{ht})$$

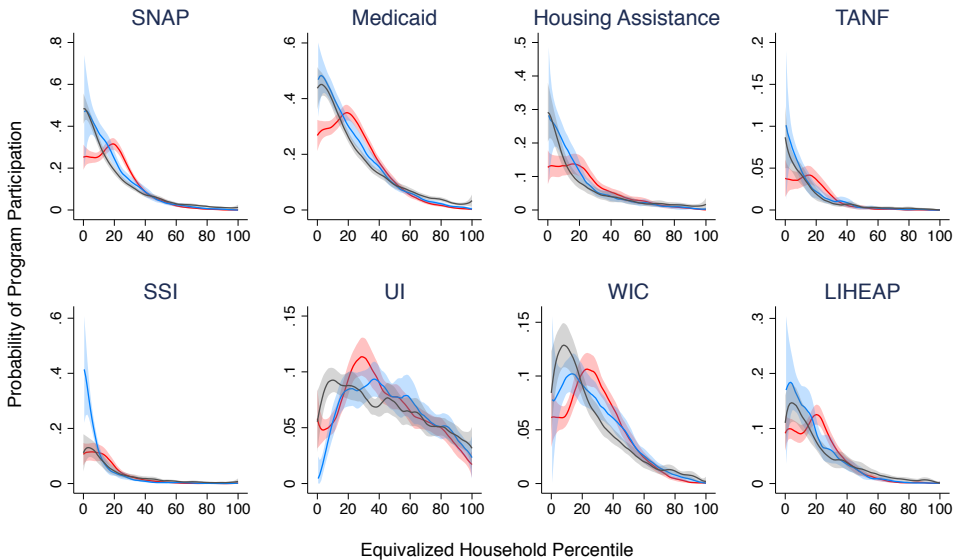
Individual lifetime earnings rank

$$\bar{r}_i = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{it}^*\right)$$

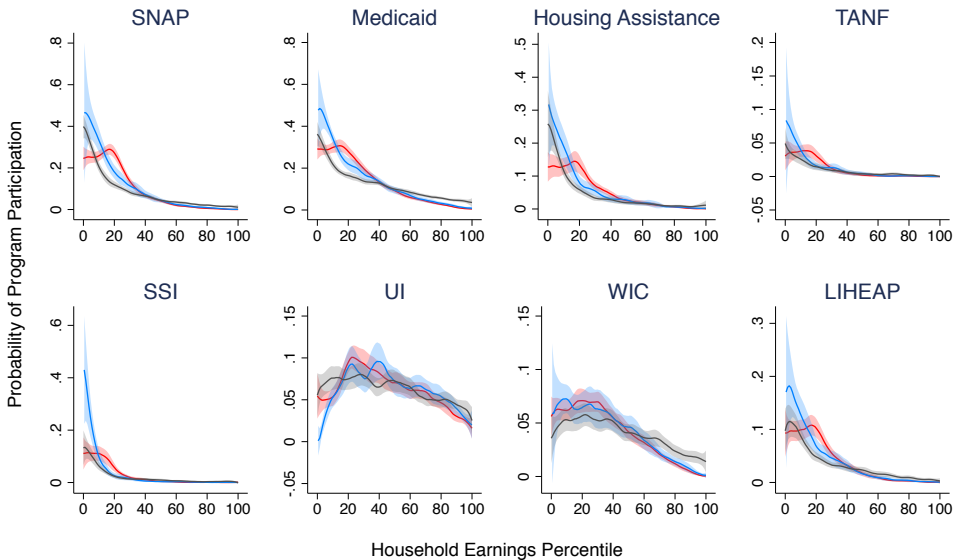
Household lifetime earnings rank

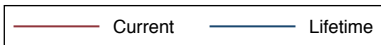
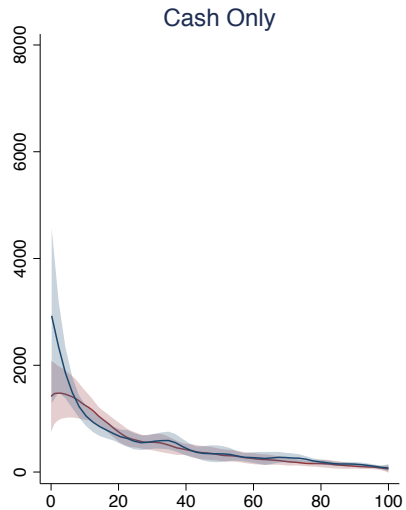
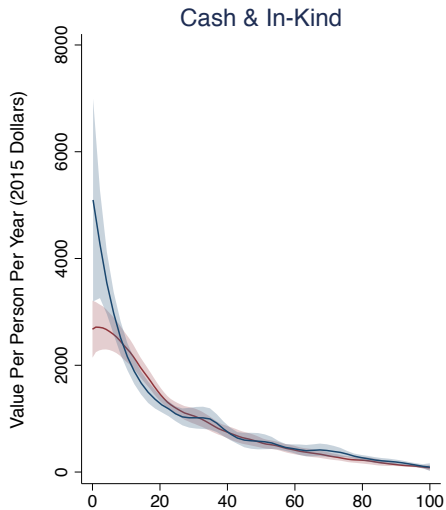
$$\bar{r}_h = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{ht}^*\right)$$

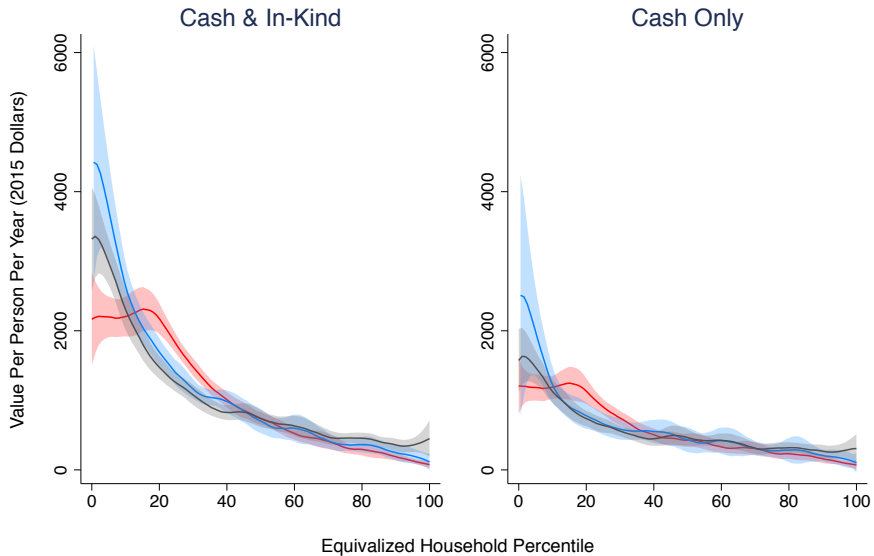




— Current Earnings — Lifetime Earnings — Current Consumption





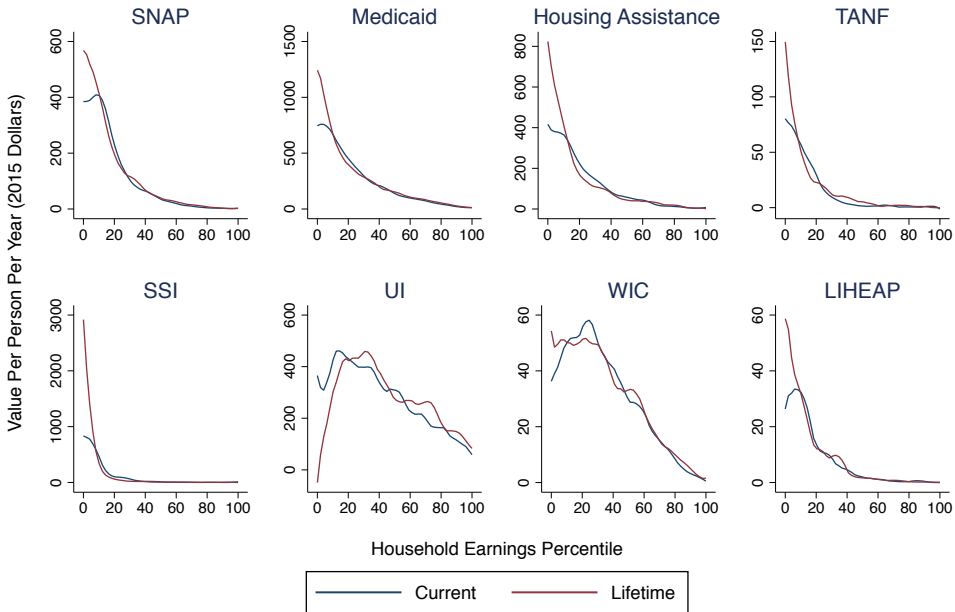


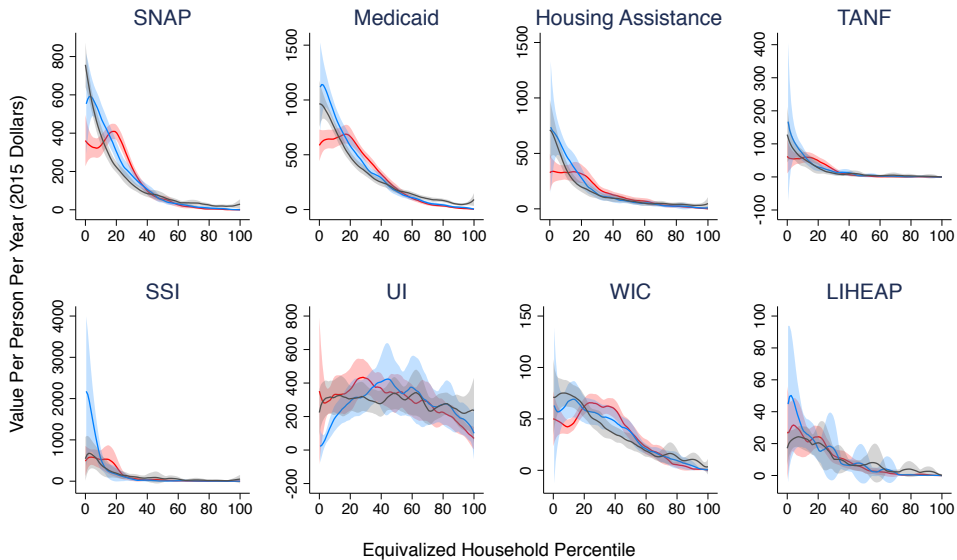
— Current Earnings — Lifetime Earnings — Current Consumption



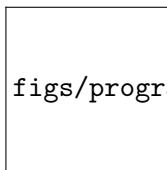
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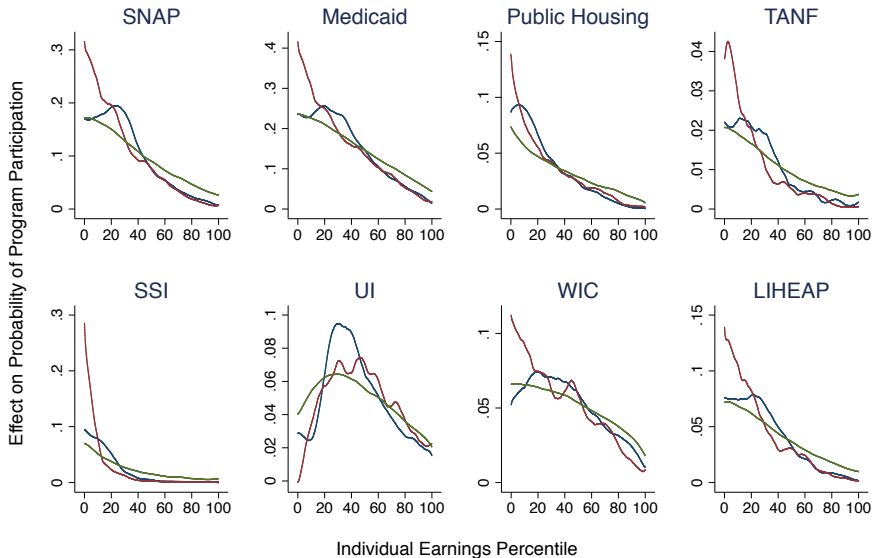


— Current Earnings — Lifetime Earnings — Current Consumption

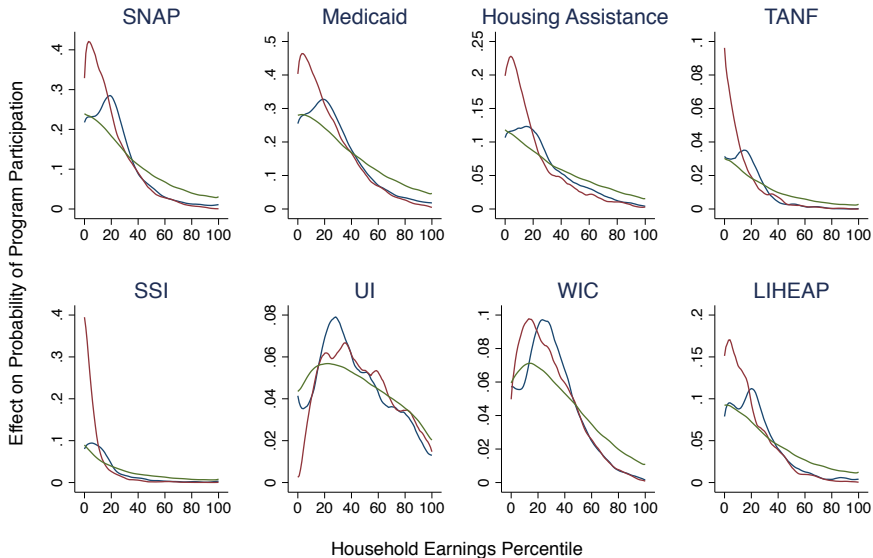


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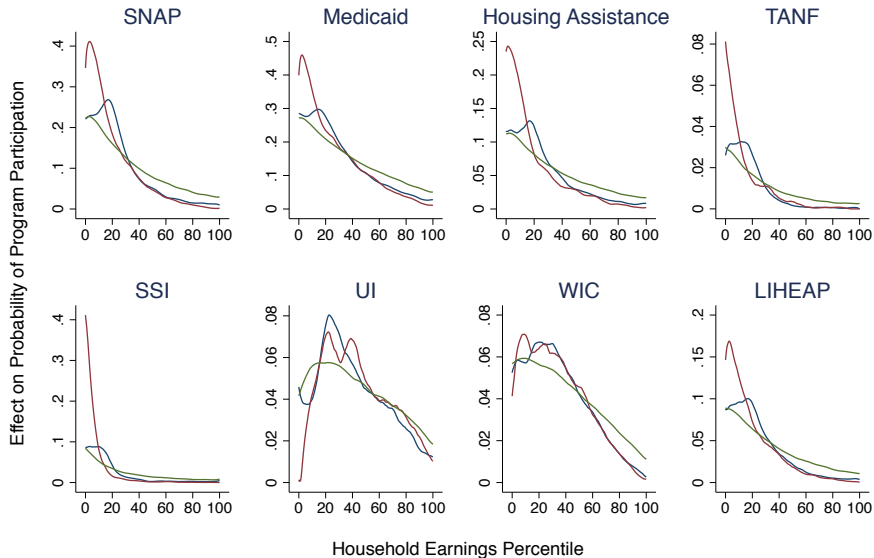
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— Current, Actual — Lifetime, Actual — Lifetime, Counterfactual



— Current, Actual — Lifetime, Actual — Lifetime, Counterfactual



— Current, Actual — Lifetime, Actual — Lifetime, Counterfactual

